

Photoplethysmography in the assessment of venous insufficiency

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Photoplethysmography has been used simultaneously with ambulatory venous pressure measurements to study 32 patients. Of the 62 limbs studied, 24 were normal, 20 had primary varicose veins, and 18 had deep valvular incompetence. The results indicate that the photoplethysmographic refilling time can provide a good separation between the three groups when the test is performed in the standing, but not in the sitting, position. However, the test is qualitative because we found a lack of correlation between the short refilling times in limbs with deep venous insufficiency and the ambulatory venous pressure. (J VASC SURG 1987;5:405-12.)

Authors have suggested that photoplethysmography can be used in the diagnosis and assessment of venous insufficiency and also for follow-up of patients who have undergone venous reconstruction.^{1,2} The purpose of our study was to determine whether photoplethysmography could differentiate between normal limbs, limbs with primary varicose veins, and limbs with deep venous insufficiency and whether it could provide a quantitative measure of the degree of insufficiency. In addition, we attempted to determine whether the measurements made with the patient standing would give better information than when made in the sitting position, which had been used in previous studies.^{1,2}

MATERIAL AND METHODS

Forty-two patients (62 limbs) who had venous problems have been studied by directional Doppler ultrasound, ambulatory venous pressure (AVP) measurements, and photoplethysmography. On the basis of directional Doppler ultrasound and AVPs³ the limbs studied were classified as (1) normal; (2) having superficial venous valvular incompetence only (primary varicose veins); and (3) having deep venous valvular incompetence (deep venous insufficiency). Limbs were considered to be normal when the AVP (lowest mean pressure during tiptoe exercise) was

less than 40 mm Hg and no reflux was detectable in the superficial or deep veins with the use of directional Doppler ultrasound as described by Shull et al.³ Limbs were considered to have primary varicose veins if reflux was detected in the long or short saphenous vein but not in the popliteal vein^{3,4} and the AVP was less than 40 mm Hg when a below-knee or thigh cuff (2.5 cm wide) or both were applied to occlude the superficial veins.³ Limbs were considered to have deep venous insufficiency if there was reflux detected in the popliteal vein³ and if they had an AVP greater than 40 mm Hg even when the superficial veins were occluded by cuffs. All patients had maximal venous outflow (MVO) measured by strain-gauge plethysmography.⁵ Patients with an MVO of less than 65 ml/100 ml/min indicating outflow obstruction have been excluded. In our hands, the normal range is 65 to 140 ml/100 ml/min.⁵

The detection of reflux by directional Doppler ultrasound

Saphenofemoral junction.⁴ The patient was examined in the standing position, facing the examiner while holding onto a rail or table to ensure complete immobility and to avoid muscular contractions in the leg. The probe of a directional Doppler ultrasonic blood velocity detector was placed over the common femoral vein at a 45-degree angle to the horizontal. The common femoral vein was easily found just medial to the artery. Confirmation that the probe was correctly positioned was obtained by manual compression of the thigh or calf—a maneuver that produces a marked increase in the blood velocity signal towards the heart. Calf compression was re-

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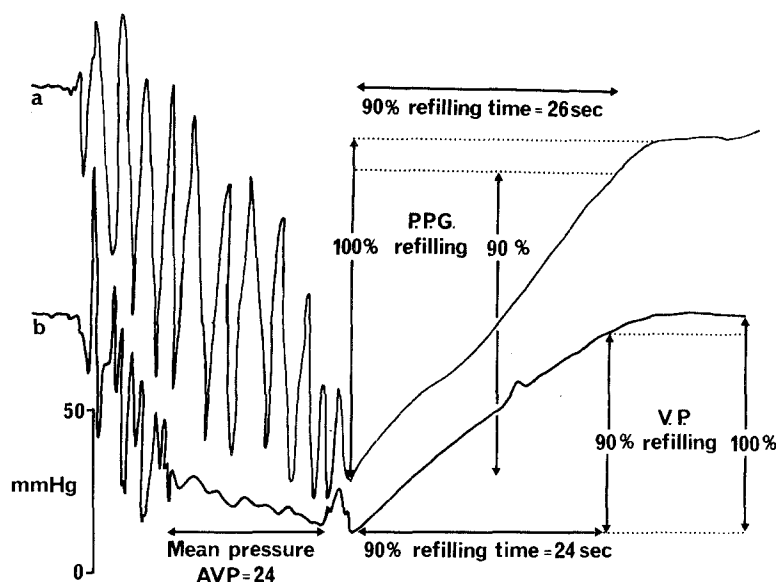


Fig. 1. Simultaneous photoplethysmographic (a) and venous pressure (b) recordings obtained in the standing position. Method of calculating 90% recovery time is shown (AVP = ambulatory venous pressure).

peated, and the position of the probe was adjusted to obtain the maximum signal, ensuring that the ultrasonic beam was at the same plane as the axial stream. The patient was then asked to cough and do a Valsalva maneuver. Subsequently, the thigh or calf was compressed manually, producing a further augmented signal of flow towards the heart, and the pressure was released suddenly. Absence of detectable flow during the cough and Valsalva maneuver and on sudden release of the thigh or calf compression meant that there was no reflux; this was characteristic of normal limbs. In contrast, a signal lasting longer than 0.5 seconds (usually 1 to 4 seconds) indicating flow away from the heart suggested reflux.^{4,6} In the presence of reflux, the thigh or calf was again compressed, and the pressure was suddenly released while the long saphenous vein was occluded by manual compression or by a rubber tube placed around the thigh 5 to 10 cm below the saphenofemoral junction. Abolition of the reflux meant that the latter was the result of an incompetent saphenofemoral junction. Persistence of the reflux despite compression of the long saphenous vein indicated that it was in the femoral vein.

Saphenopopliteal junction and popliteal vein.^{3,4} The patient was examined standing, facing away from the examiner, and holding onto a railing or table. The knee on the side to be examined was flexed slightly (heel on the ground with foot slightly forward), so as to relax the popliteal fossa. The

weight of the body was kept on the opposite limb. This position ensured that the muscles of the leg to be examined were not contracting. The Doppler probe was then positioned in the popliteal fossa, with light pressure placed on the skin at the level of the skin crease, and at an angle of 45 degrees to the horizontal so that the signal from the popliteal artery, in addition to the signal from the popliteal vein, was audible. Augmentation of the venous signal by calf compression and fine adjustment of the probe position ensured an optimal venous signal. The calf was then compressed manually and venous flow in the direction of the heart was detected. Absence of flow on sudden release of the calf compression meant that there was no reflux and was characteristic of a normal saphenofemoral junction and popliteal vein. The presence of flow distally indicated reflux into either the short saphenous (incompetent saphenopopliteal junction) or popliteal vein. In the presence of reflux, the calf compression with sudden release was repeated with the addition of finger compression, by an assistant or nurse, of the short saphenous vein just distal to the probe. If reflux persisted despite digital compression of the short saphenous vein, it was inferred that the reflux was in the popliteal vein itself. It has already been demonstrated that popliteal vein reflux can be detected by this method with a sensitivity of 100% and specificity of 92%.⁴

The measurement of AVPs.³ A 23-gauge butterfly needle was inserted into a vein on the dorsum

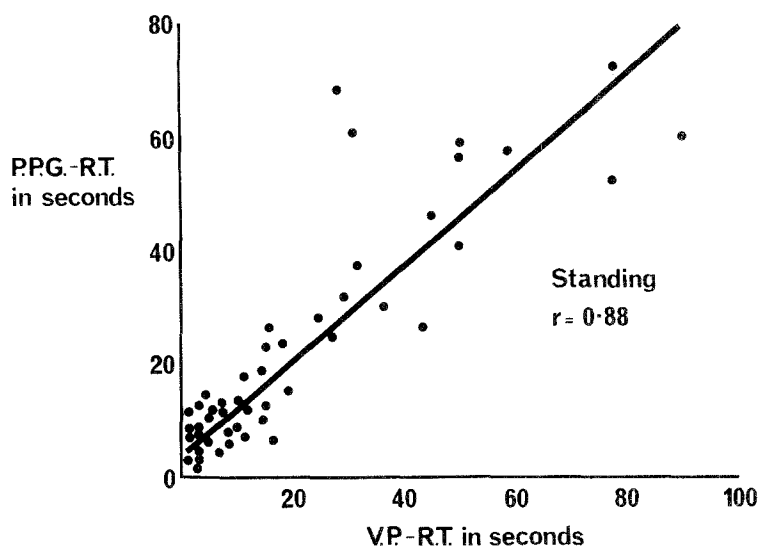


Fig. 2. Relationship between photoplethysmographic refilling time (P.P.G.-R.T.) and the venous pressure refilling time (V.P.-R.T.) obtained simultaneously in the standing position.

of the foot and was connected through a Gould transducer (Gould Electronics Ltd., Ilford, Essex, U.K.) and an amplifier to a potentiometric pen recorder. The venous pressure was first measured with the patient standing still. Then, the patient was asked to perform a tiptoe exercise for 30 seconds at a rate of one heel-raising movement per second. Throughout these measurements, the patient was holding onto a railing ensuring the absence of muscular contractions at rest and a smooth recording during the refilling time, immediately after the exercise. The exercise was repeated after inflation of a 2.5 cm pneumatic cuff at the ankle to a pressure of 120 mm Hg and a second time after inflation of a similar cuff at the thigh, to a pressure of 180 mm Hg. At these pressures, the narrow cuff used occludes the superficial veins only, but not the deep veins, as shown by previous phlebographic studies.^{7,8} The patient was then studied in the sitting position (on a couch) with the foot off the ground. The exercise consisted of active plantar flexion of the foot at the rate of one per second for 30 seconds. Each bout of exercise was repeated three times, without and with inflation of the ankle cuff.

Our apparatus has a switch that changes the time constant to three seconds; this was switched on during the exercise to provide a mean pressure but it was not used during the last few seconds of the exercise and the recovery time. All recordings were repeated three times. From each recording the AVP was determined as the lowest mean pressure achieved during the exercise. In addition, the venous pressure

refilling time (VP-RT) was determined. The latter was defined as the time in seconds from the end of exercise to the point the pressure had returned to 90% of the resting pressure (Fig. 1).

Photoplethysmography. Photoplethysmographic recordings were made simultaneously with the AVP recordings by means of another channel of the same pen recorder. The photoplethysmographic probe (Photopulse PH77, MedaSonics, Mountain View, Calif.) was affixed to the skin of the dorsum of the foot with transparent tape that had adhesive on both sides. The photoplethysmographic refilling time (PPG-RT) was determined from the photoplethysmographic recording. This was defined as the time in seconds from the end of exercise to the point where the recording reached 90% of the distance between the plateau during exercise and the eventual postexercise plateau measured on the vertical axis (Fig. 1).

RESULTS

Of the 62 limbs studied, 24 were normal, 20 had primary varicose veins only, and 18 had deep venous insufficiency. This classification was based on the standing AVPs and the presence or absence of reflux in the superficial and deep veins according to the criteria postulated (see Methods section). The normal limbs belonged to patients who had either primary varicose veins or deep venous insufficiency confined to the opposite limb.

The correlation between PPG-RT and VP-RT obtained simultaneously from the same leg in the

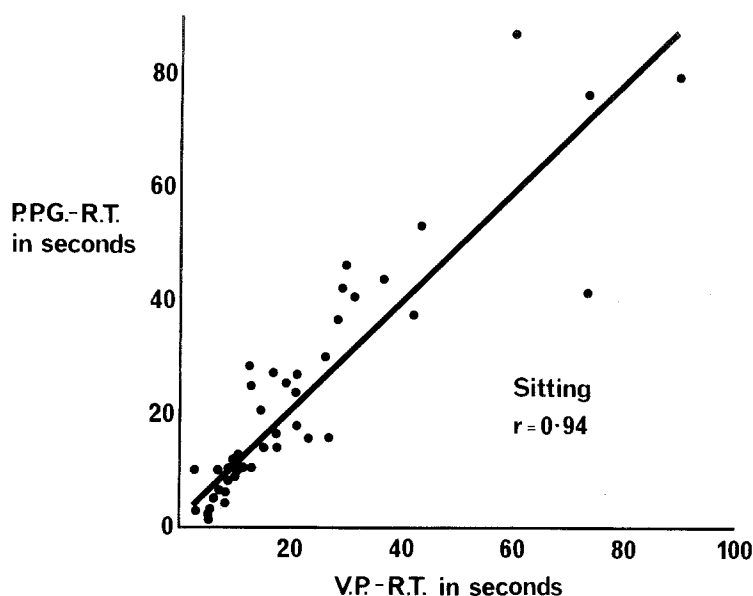


Fig. 3. Relationship between photoplethysmographic refilling time (*P.P.G.-R.T.*) and the venous pressure refilling time (*V.P.-R.T.*) obtained simultaneously in the sitting position.

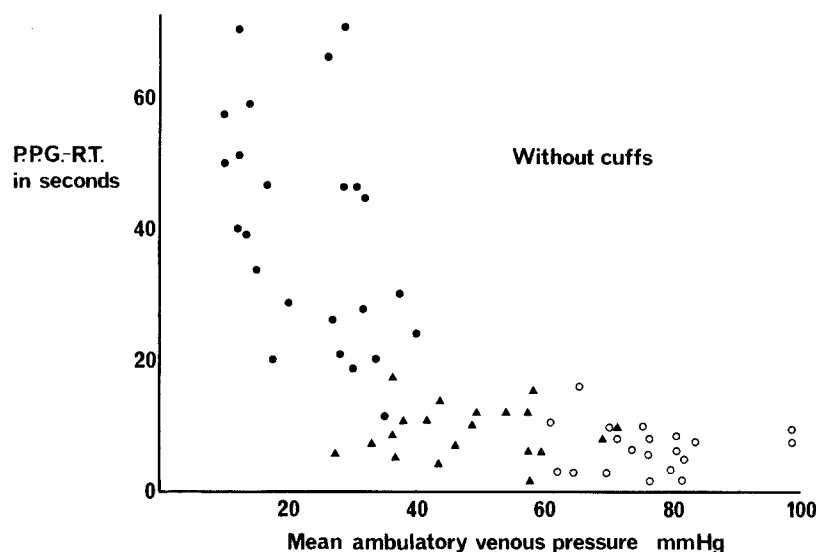


Fig. 4. Relationship between photoplethysmographic refilling time (*P.P.G.-R.T.*) and mean ambulatory venous pressure obtained in the standing position with the cuff deflated (*closed circles* = normal limb; *triangles* = superficial venous incompetence only; *open circles* = deep venous incompetence).

standing and sitting positions (no cuffs) is shown in Figs. 2 and 3. A linear relationship was found between the PPG-RT and the VP-RT, both in the standing ($r = 0.88$) and sitting ($r = 0.94$) positions.

Because the severity of venous insufficiency can be expressed by the mean AVP in the erect position, the latter was plotted against the PPG-RT without (Fig. 4) and with (Fig. 5) cuffs occluding the su-

perficial veins. These plots demonstrate the distribution of the three groups of limbs studied and the effect of the cuffs. There was an extensive overlap between the PPG-RT of those limbs with primary varicose veins (superficial reflux) and the PPG-RT of those limbs with deep venous incompetence when the ankle cuff was not inflated (Fig. 4). After the reflux in the superficial veins was abolished by inflating the cuff, a better separation of the limbs with

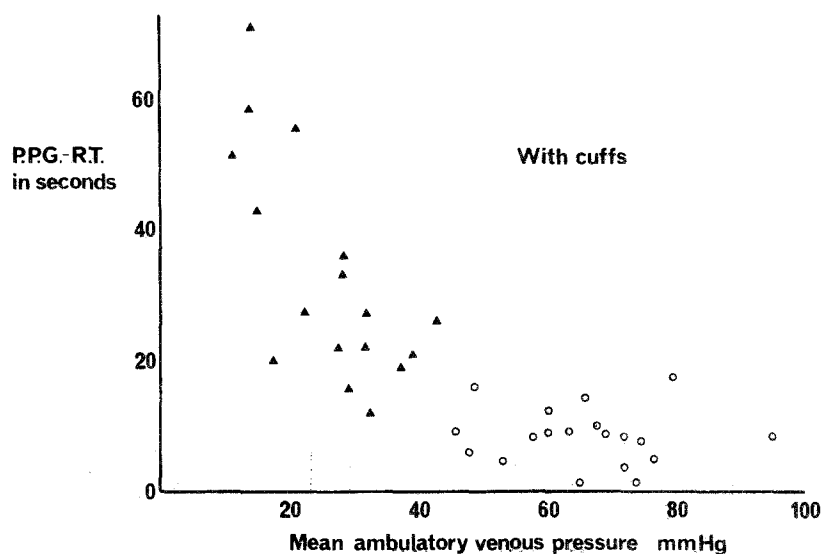


Fig. 5. Relationship between photoplethysmographic refilling time (P.P.G.-R.T.) and mean ambulatory venous pressure obtained in the standing position with cuff inflated (*triangles* = superficial venous incompetence only; *open circles* = deep venous incompetence). Normal limbs are not shown; the inflated cuff did not alter their position as shown in Fig. 4.

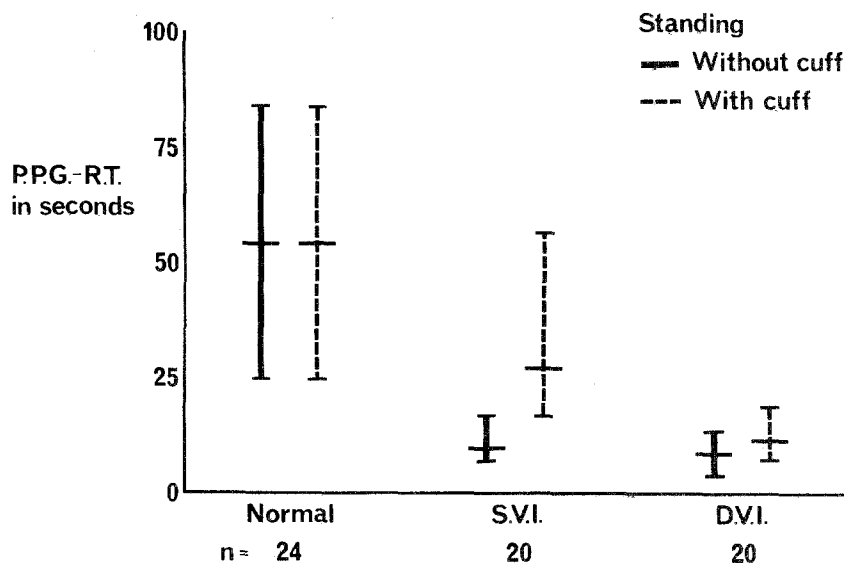


Fig. 6. Median and 95% tolerance levels of the photoplethysmographic refilling time (P.P.G.-R.T.) in the three groups of limbs studied in the standing position. *Interrupted lines* represent the 95% tolerance levels of the PPG-RT when ankle cuff was inflated. S.V.I. = superficial venous insufficiency; D.V.I. = deep venous insufficiency.

primary varicose veins from those with deep venous reflux was effected (Fig. 5). The PPG-RT was increased to more than 18 seconds in all but two limbs with superficial venous incompetence; it remained less than 18 seconds in all limbs with deep venous incompetence.

The separation between the groups is more clearly shown by using the median and 95% tolerance levels

(i.e., the range of 95% of the data) of the PPG-RT in the standing (Fig. 6) and sitting (Fig. 7) positions. Fig. 6 indicates that normal limbs can be identified by a PPG-RT more than 18 seconds when tested without a cuff, in contrast to limbs in both groups of superficial and deep venous insufficiency that have a PPG-RT less than 18 seconds. By applying a cuff and repeating the tests, limbs in the group with su-

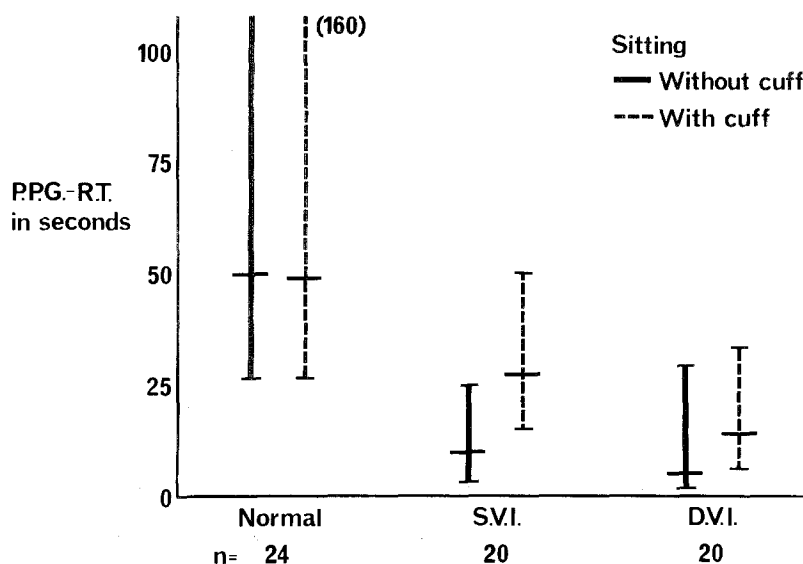


Fig. 7. Median and 95% tolerance levels of the photoplethysmographic refilling time (P.P.G.-R.T.) in three groups studied in the sitting position. *Interrupted lines* represent the 95% tolerance levels of the PPG-RT when ankle cuff was inflated.

perforial venous insufficiency will have a PPG-RT of 18 seconds or longer. This will not happen to limbs in the group with deep venous insufficiency. In the sitting position (Fig. 7) the application of the cuff will not produce such a separation between these two groups.

Finally, Figs. 4 and 5 demonstrate that, in limbs with deep venous insufficiency, the PPG-RT is a poor measure of the severity of the condition, because although it may be short (e.g., 10 seconds), it cannot indicate what the AVP is. The AVP can range from 45 to 100 mm Hg.

DISCUSSION

Photoplethysmography is a noninvasive technique that can detect changes in the blood content of the dermis. Although skin is essentially opaque to light, slight light transmission and backscattering in the range of the visible and infrared spectrum does occur. In practice, a probe consisting of a light source and a light-sensitive diode is positioned on the skin. Changes in the number of red cells in the dermis affect the backscatter of light and are detected by the light-sensitive diode.

The first use of photoplethysmography was reported in the 1930s.^{9,10} Subsequently, Hertzman and Dillon¹¹ gave it the title of photoelectric plethysmography and used it to measure blood volume changes induced in the skin by exposure to cold and by exercise in normal people and in patients with

peripheral arterial disease. They also reported the circuitry required to distinguish the changes of blood flow and blood volume in the dermis.¹² In addition, he demonstrated that in contrast to the blood flow signal (alternating current signal) synchronous to the pulse, the blood volume signal (direct current signal) changes slowly, depending on changes in the dermal blood volume. Such changes are produced by alterations in venous pressure caused by changes in limb position or proximal venous occlusion. Photoplethysmography has found little application because of the inability to calibrate the signal, unless one is using time measurements such as the postexercise refilling time used in the present study.

Exercise in the normal subject results in a prompt reduction in venous pressure and dermal blood content, as demonstrated by the simultaneous recording of venous pressure and photoplethysmography (Fig. 1). The time taken for postexercise refilling to occur depends on how much the veins have been emptied during the exercise, the competence of the valves in the superficial and deep veins, and the arterial inflow. In the normal limb, postexercise refilling depends entirely on the arterial inflow, and when standing, it ranges from 19 to 80 seconds. In the presence of venous incompetence, refilling is rapid (less than 18 seconds) because of the reflux in the incompetent venous channels, regardless of whether this is occurring down the superficial veins, the deep veins, or both.

The PPG-RT determined in the standing position separates limbs without valvular incompetence from those with valvular incompetence (Fig. 6). This separation occurs even when the AVP falls within the normal range, as it may in patients with relatively mild superficial incompetence. In those patients with PPG-RT of less than 18 seconds the test is repeated with the ankle cuff inflated. A prolongation of the refilling time to more than 18 seconds can distinguish superficial from deep incompetence correctly in 95% of limbs. This separation is not so good in the sitting position.

The AVP is a good indicator of the severity of venous insufficiency. Unfortunately, a linear relationship does not exist between the PPG-RT and the mean AVP (Figs. 4 and 5). A refilling time more than 18 seconds is normal and means that the AVP would be less than 40 mm Hg. A refilling time less than 18 seconds is indicative of an abnormal AVP that could have any value between 40 and 100 mm Hg (Figs. 4 and 5). Consequently, PPG-RT cannot grade the severity of venous insufficiency; it can only detect its presence. This is contrary to the conclusions of previous work by Norris, Beyrau, and Barnes¹³ who attempted to calibrate the photoplethysmographic device. Although they did the test with the patient in the standing position and showed an excellent correlation between AVP refilling time (AVP-RT) and PPG-RT they did not attempt to correlate refilling time with actual AVP.

In our hands, the principal use of photoplethysmography is as a screening method for patients with venous problems who have already been shown not to have outflow obstruction ($MVO > 65 \text{ ml}/100 \text{ ml/min}$). It is an easily performed noninvasive test that permits the classification of patients into one of three clinically important groups: (1) normal valvular function; (2) superficial valvular incompetence only; or (3) deep valvular incompetence (with or without an element of superficial valvular incompetence).

The diagnostic flow chart we use for the screening of patients employs two diagnostic techniques: Doppler ultrasound and photoplethysmography (Fig. 8). Limbs with a PPG-RT more than 18 seconds are considered normal. In limbs with a PPG-RT less than 18 seconds the exercise is repeated with the ankle cuff inflated. Prolongation of the PPG-RT to more than 18 seconds means superficial venous incompetence only (primary varicose veins), and this can be repeated with the cuff below or above the knee. Once the cuff is placed above the level of the most distal site of deep-to-superficial reflux, the PPG-RT reverts to less than 18 seconds. Thus, this

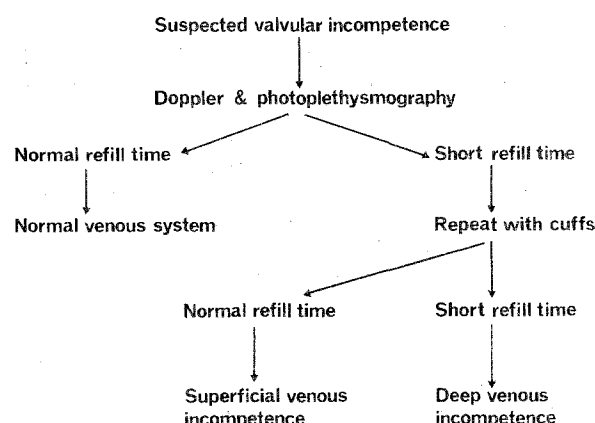


Fig. 8. Diagnostic flow chart indicates procedure followed for screening of patients with suspected chronic venous problems.

technique can determine the most distal site of deep-to-superficial reflux.

Doppler ultrasound will confirm the presence of reflux at the popliteal and more proximal sites. A PPG-RT of less than 18 seconds despite the application of the pneumatic cuff at the ankle indicates incompetence of the valves in the deep veins, including the tibial. This is an indirect approach to the detection of reflux. A direct method would be to use duplex scanning, which enables the detection of reflux in individual veins that have been anatomically identified. If duplex scanning were to prove practical and accurate in detecting reflux in the tibial veins, then photoplethysmography may become unnecessary. Until such time photoplethysmography will remain the method of choice.

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